

Arjuna JEE 2025

Maths

DPP: 9

Trigonometric Identities

Q1 Domain and range of $f(x) = 3 \sin 4x$ are

- (A) $x \in R, f(x) \in [-3, 3]$
 (B) $x \in R^f, f(x) \in (-3, 3)$
 (C) $x \in R^-, f(x) \in (-\infty, \infty)$
 (D) None of these

Q2 The maximum value of $3 \cos \theta + 4 \sin \theta$ equal to:

- (A) 3 (B) 4
 (C) 5 (D) None of these

Q3 If m is the minimum value of $f(x) = 3 \sin x + 5$ and n is the maximum value of $g(x) = 3 - 2 \sin x$, then $(m + n + 2)$ is equal to

- (A) 11 (B) 9
 (C) 8 (D) 10

Q4 The maximum value of $1 + \sin\left(\frac{\pi}{4} + \theta\right) + 2 \cos\left(\frac{\pi}{4} - \theta\right)$ for real values of θ is

- (A) 3 (B) 5
 (C) 4 (D) 2

Q5 The range of $f(\theta) = 3 \cos^2 \theta - 8\sqrt{3} \cos \theta \cdot \sin \theta + 5 \sin^2 \theta - 7$ is given by

- (A) $[-7, 7]$ (B) $[-10, 4]$
 (C) $[-4, 7]$ (D) $[-10, 7]$

Q6 The least value of $\cos^2 \theta - 6 \sin \theta \cos \theta + 3 \sin^2 \theta + 2$, is

- (A) $4 + \sqrt{10}$

(B) $4 - \sqrt{10}$

(C) 0

(D) none of these

Q7 The maximum value of $3 \cos \theta + 5 \sin\left(\theta - \frac{\pi}{6}\right)$ for any real value of θ is :

- (A) $\sqrt{34}$
 (B) $\sqrt{31}$
 (C) $\sqrt{19}$
 (D) $\sqrt{\frac{79}{2}}$

Q8 If $\sin x_1 + \sin x_2 + \sin x_3 + \dots + \sin x_{2021} = 2021$
 Then value of $\cos x_1 + \cos x_2 + \dots + \cos x_{2021} =$

- (A) 1
 (B) 0
 (C) π
 (D) $\frac{\pi}{2}$

Q9 If $p = \sin^2 x + \cos^4 x$, then

- (A) $\frac{3}{4} \leq p \leq 1$
 (B) $\frac{3}{16} \leq p \leq \frac{1}{4}$
 (C) $\frac{1}{4} \leq p \leq \frac{1}{2}$
 (D) None of these

Q10 The ratio of the greatest value of $2 - \cos x + \sin^2 x$ to its least value, is

- (A) $7/4$ (B) $11/4$
 (C) $13/4$ (D) none of these

Q11 The value of $\sin \theta + \cos \theta$ will be greatest when

- (A) $\theta = 30^\circ$



- (B) $\theta = 45^\circ$
 (C) $\theta = 60^\circ$
 (D) $\theta = 90^\circ$

Q12 If $4 \sin 27^\circ = \sqrt{\alpha} - \sqrt{\beta}$, then the value of $\alpha + \beta$ is

- (A) 5 (B) 3
 (C) 8 (D) 2

Q13 Minimum value of $27^{\cos x} + 81^{\sin x}$ is equal to

- (A) $\frac{2}{3\sqrt{3}}$
 (B) $\frac{2}{9\sqrt{3}}$
 (C) $\frac{4}{3\sqrt{3}}$
 (D) None of these

Q14 If the maximum value of expression $\frac{1}{5 \sec^2 \theta - \tan^2 \theta + 4 \operatorname{cosec}^2 \theta}$ is equal to $\frac{p}{q}$ (where p and q are coprime), then the value of $(p + q)$ is

- (A) 14 (B) 15
 (C) 16 (D) 18

Q15 Maximum value of $16^{\sin^2 x} \cdot 32^{\cos^2 x}$ is equal to

- (A) 8 (B) 32
 (C) 16 (D) 4



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Answer Key

Q1 (A)

Q2 (C)

Q3 (B)

Q4 (C)

Q5 (B)

Q6 (B)

Q7 (C)

Q8 (B)

Q9 (A)

Q10 (C)

Q11 (B)

Q12 (C)

Q13 (B)

Q14 (D)

Q15 (B)



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